

CONTACT

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SKILLS

Concurrent Data Structures	7+ yrs
High Performance Computing	6+ yrs
Program Analysis	1+ yrs
C++	8+ yrs
Python	7+ yrs
Java	3+ yrs
JavaScript	2+ yrs
C#	1+ yrs
SQL	1+ yrs

KENNETH LAMAR

Software Engineer - Google

WORK EXPERIENCE

Software Engineer - GPU System Software Google - Sunnyvale, CA (USA) Develop Baseboard Management Controller (BMC) software. Maintain reliability monitoring services, kernel rollouts, firmware and driver upgrades, cooling, and power management for emerging GPU platforms. Tools: C, C++, Python, Linux kernel development	2025 - Present
Post Doctoral Scholar University of Central Florida - Orlando, FL (USA) Finalized graduate research projects and trained new graduate and undergraduate students.	2024 - 2025
Computing Graduate Student Intern Lawrence Livermore National Laboratory - Livermore, CA (USA) Evaluated code quality metrics to improve maintainability. Created ROSE LCOM Tools to measure class cohesion and a tool to measure code churn. Tools: C++, Python, Ada, ROSE compiler, git churn, gprof, static analysis	Summer 2023
Applications Developer Internship MVP Sports Clubs - Orlando, FL (USA) Developed customer touchpoint system, guest check-in alert, customer risk factor identification, iOS and Android apps, and API integrations. Tools: ASP.NET, SQL, C#, JavaScript, Java, Swift	2017

EDUCATION

Doctor of Philosophy - Computer Science University of Central Florida - Orlando, FL (USA) Advisor: Dr. Damian Dechev Dissertation Topic: Concurrent data structures & HPC scheduling	2018 - 2024
Master of Science - Computer Science University of Central Florida - Orlando, FL (USA) Masters along the way	2018 - 2023
Bachelor of Science - Computer Science University of Central Florida - Orlando, FL (USA) Minor in Mathematics	2014 - 2017
Associate of Arts Daytona State College - Daytona Beach, FL (USA)	2011 - 2014

PUBLICATIONS

Predicting HPC Job Run time with Realistic Data Using Application Input Parameters
29th Annual IEEE High Performance Extreme Computing, September 2025

HPEC 2025

ROSE LCOM Tools

ACM International Conference on the Foundations of Software Engineering, June 2025

FSE 2025

Evaluating HPC Job Run Time Predictions Using Application Input Parameters

17th ACM International Conference on Distributed and Event-Based Systems, June 2023

DEBS 2023

Metrics for Packing Efficiency and Fairness of HPC Cluster Batch Job Scheduling

IEEE 34th International Symposium on Computer Architecture and High Performance Computing, November 2022

Secondary author

SBAC-PAD 2022

Backfilling HPC Jobs with a Multimodal-Aware Predictor

Workshop on Monitoring and Analysis for HPC Systems Plus Applications, September 2021

Co-located with CLUSTER

HPCMASPA 2021

PMap: A Non-volatile Lock-free Hash Map with Open Addressing

2021 IEEE 10th Non-Volatile Memory Systems and Applications Symposium, August 2021

NVMSA 2021

Lock-free transactional vector

11th International Workshop on Programming Models and Applications for Multicores and Manycores, February 2020

Co-located with PPOPP

PMAM 2020

An Efficient Latch-free Database Index Based on Multi-dimensional Lists

37th IEEE International Performance Computing and Communications Conference, November 2018

IPCCC 2018

PRESENTATIONS

Tilt-Shift Rendering Using a Thin Lens Model

Student Presentation - UCF - Orlando, FL (USA)

Provided an explanation of and developed an interactive web demo fully simulating a tilt-shift lens using real-time ray tracing.

Tools: TWGL, D3.js, reveal.js, WebGL shaders

Apr 2022

A Persistent Hash Map for Graph Processing Workloads and a Methodology for Persistent Transactional Data Structures

CppCon 2021 - Aurora, CO (USA)

Presented my work on PMap, a persistent hash map design.

Sep 2021

RacerD: Compositional Static Race Detection

Apr 2020

Student Presentation - UCF - Orlando, FL (USA)

Presented on RacerD, a static analysis tool to detect data races, designed by Facebook. Ran on four popular Android apps and 9 toy programs and identified several potential and real data races.

Tools: RacerD

PROJECTS

Feature Creeps

2024-Present

Tools: TypeScript, WebRTC, WebLLM, OpenAI API, Kokoro.js, VITS-web, Astro, Static site

Developed the backend for a Jackbox Party Pack-style party game where multiplayer communication, LLM, and TTS systems all run server-free

CONVUL Reimplementation

Apr 2021

Tools: Intel PIN, C++, CONVUL

Recreated CONVUL, a concurrency vulnerability detector using dynamic analysis, as a student project, since the original design did not have source code available.

24-Player Mario Kart Split Screen Multiplayer

2017 - 2026

Tools: WinAPI, libusb, ViGEM, Batch scripting, Dolphin, VMWare, dnsmasq

Multi-instance workflow and tooling for massively multiplayer splitscreen. Wrote a custom controller driver, a window tiling tool, and documentation for setup and usage. Supports many other games too.

TEACHING

Graduate Teaching Assistant

Spring 2021

COP 3402 - Systems Software - UCF

Instructor: Euripides Montagne

Graduate Teaching Assistant

Spring 2020

COP 4520 - Multicore Programming - UCF

Instructor: Damian Dechev

Graduate Teaching Assistant

Fall 2019

CAP 4102 - Web Design and User Experience - UCF

Instructor: Reza Aria

Graduate Teaching Assistant

Spring 2019

CIS 3360 - Security in Computing - UCF

Instructor: Joshua Lazar

Graduate Teaching Assistant

Fall 2018

CIS 3360 - Security in Computing - UCF

Instructor: Michael McAlpin

AWARDS

- 2018 UCF College of Graduate Studies Presentation Fellowship
- President's List - Daytona State College - 7 times between 2011-2015